Practical 4: Performing Data Analysis Using Calculations in Tableau.

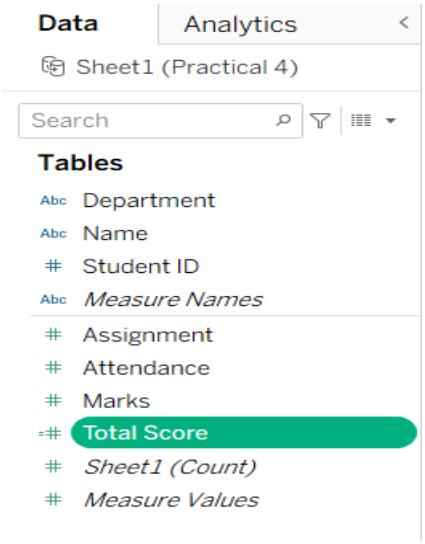
Aim :
To create calculated fields, perform row-level and aggregate calculations, implement Level of Detail (LOD) expressions, create parameters, and use calculations for data analysis.
a) Create basic calculated fields using formulas
b) Perform row-level calculations
c) Perform aggregate-level calculations
d) Implement FIXED Level of Detail (LOD) calculations
e) Use INCLUDE and EXCLUDE LOD expressions
f) Create and use parameters for dynamic analysis
g) Apply calculations to fix data issues and enhance insights

(A)Aim:- Create Basic Calculated Fields Using Formulas.

Step 1:-
Open Tableau and connect to "**Practical_2.csv".**

Step 2:-
Go to **Analysis → Create Calculated Field**.

<u>Example 1: Total Score</u>
Name:
Total Score
Formula:
[Marks] + [Assignment]
Click **OK**.



Step 2: Display the Result

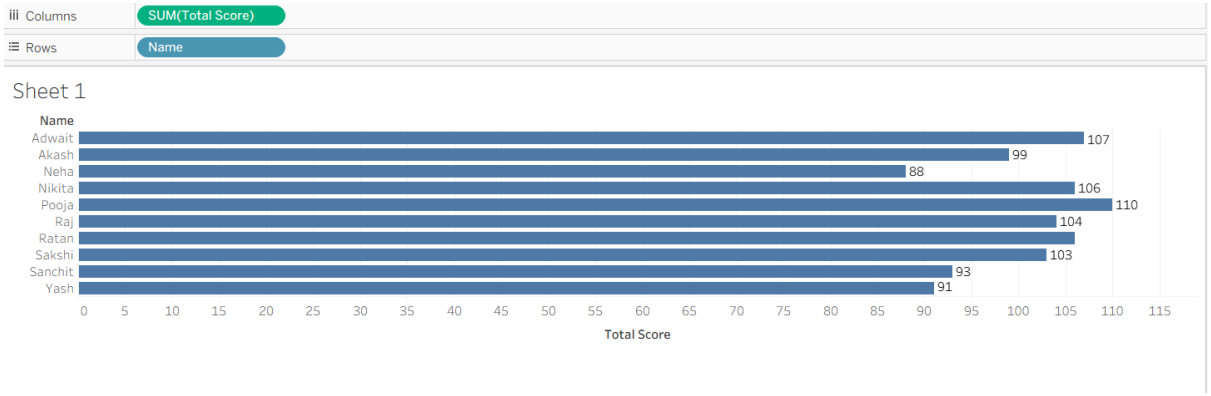
Open **Sheet 1**

Drag:

- **Name** → Rows
- **Total Score** → Text (Marks Card)

Drag

- **Name** → Rows
- **Total Score** → Columns



Example 2: Pass/Fail

Name: Result

Formula:

IF [Marks] >= 40 THEN

"Pass"

ELSE

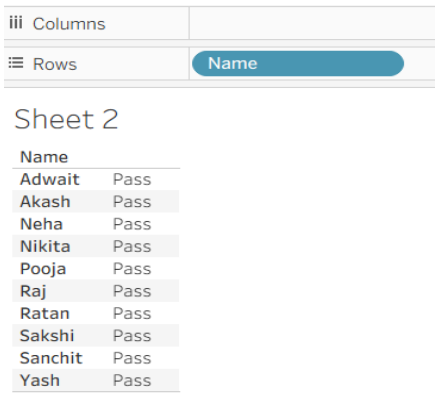
"Fail"

END

Display the Result:

Drag

- **Name → Rows**
- **Result → Text**



(B)Aim:- Perform Row-Level Calculations.

A row-level calculation is performed **for each record**.

Example:
Percentage after adding assignment marks:
Percentage=120 (Marks + Assignment)×100
([Marks] + [Assignment]) / 120 * 100
Each student gets their own percentage.

Result :
Drag:

- **Name** → Rows
- **Percentage** → Text (Marks Card)

Student 1: Amit
Marks = **75**
Assignment = **15**
75 + 15 = 90
90 / 120 = 0.75
0.75 × 100 = 75%
Result: 75%



Sheet 3

Name	
Adwait	89.17
Akash	82.50
Neha	73.33
Nikita	88.33
Pooja	91.67
Raj	86.67
Ratan	88.33
Sakshi	85.83
Sanchit	77.50
Yash	75.83

(C)Aim:- Perform Aggregate-Level Calculations.

Aggregate calculations summarize data.

Example 1

Average Marks

- Method 1: Show Overall Average

It means it shows total no.of student's average marks.

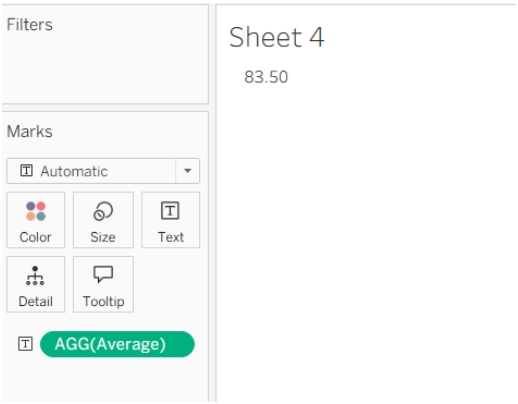
85+72+91+65+78+55+35+96+67+81/10

AVG([Marks])

Result :

Drag **Average Marks** → **Text** on the Marks card.

(Displayed total no .of student's average marks.)

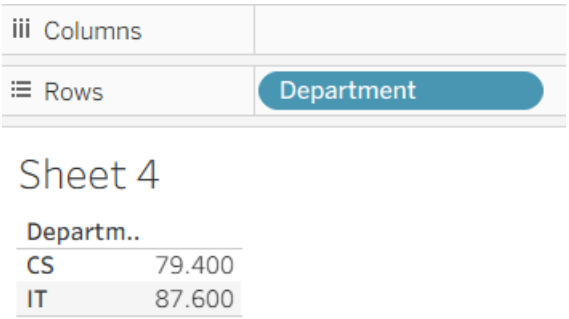


- Method 2: Department-wise Average

Drag:

Department → Rows

Average Marks → Text



Example 2:

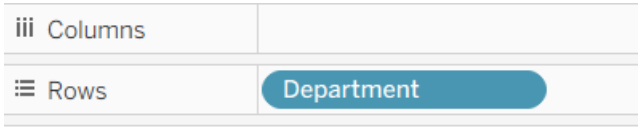
Total Marks
SUM([Marks])
Result :
Drag

- **Total Marks** → Text

(Student wise total marks)
Drag

- **Department** → Rows
- **Total Marks** → Text

(Department wise total marks)



Sheet 4

Departm..	
CS	397
IT	438

Example 3:

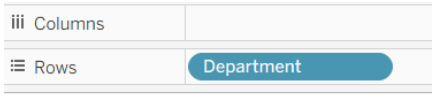
Highest Marks
MAX([Marks])
Drag

- **Highest Marks** → Text

(Department-wise Highest Marks)

Drag

- **Department** → Rows
- **Highest Marks** → Text



Sheet 5

Departm..	
CS	87
IT	92

(D)Aim:- Implement FIXED Level of Detail (LOD).

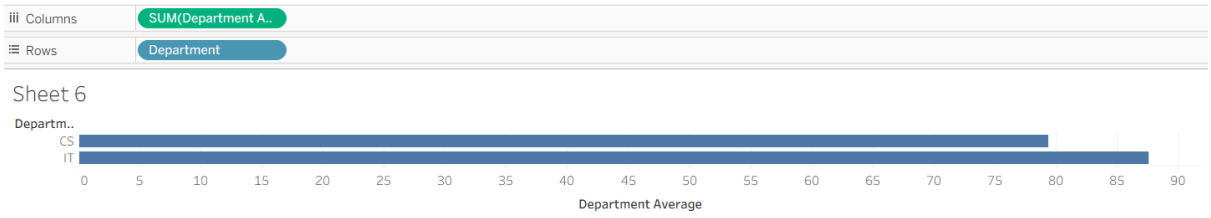
- **Definition:-**
  A **FIXED Level of Detail (LOD)** expression calculates a value at a specified level of detail **regardless of what is displayed in the worksheet**.

  Formula
  { FIXED [Department] : AVG([Marks]) }
  Name:
  Department Average

- **What does this mean?**
  It means
  "Calculate the average marks for every Department."
  For IT
  85+91+78+67/4
  =80.25
  For CS
  72+65+55+96+81/5
  =73.8

  Result :
  Drag → Department to rows.
  Drag → Department average to text.

  (another way)
  Drag:
  Department → Rows
  Department Average → Columns



(E)Aim:- Use INCLUDE and EXCLUDE LOD Expressions.

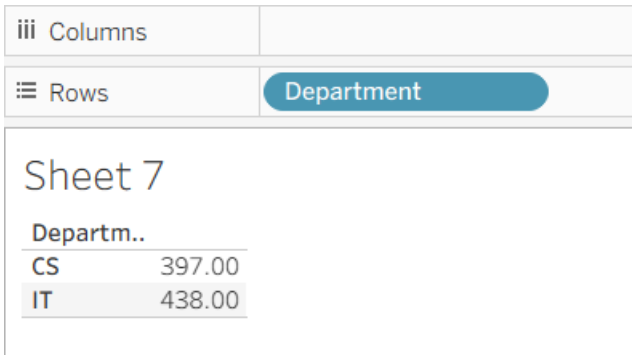
INCLUDE means
"Include an extra dimension while calculating."

INCLUDE:
Include Student Name while calculating department average.
{ INCLUDE [Name] : AVG([Marks]) }
The view only has **Department**, so Tableau aggregates these student averages again.
IT Department
Student averages:

- Amit = 80
- Neha = 70
- Ravi = 90

Department Average
80+70+903/3=451
Drag:

- **Department** → Rows
- **Include Average** → Text



EXCLUDE:
EXCLUDE removes one or more dimensions from the calculation.
Ignore Student Name and calculate only department average.

What exactly did EXCLUDE do?
Suppose your data is:

Department	Name	Marks
IT	Amit	80
IT	Neha	70
IT	Ravi	90

Your worksheet contains:

- Department
- Name

So normally Tableau sees the data like this:
IT → Amit → 80
IT → Neha → 70
IT → Ravi → 90
If you use:
AVG([Marks])
Tableau calculates for **each student** because both **Department** and **Name** are in the view.
Result:

Department	Name	AVG(Marks)
IT	Amit	80
IT	Neha	70
IT	Ravi	90

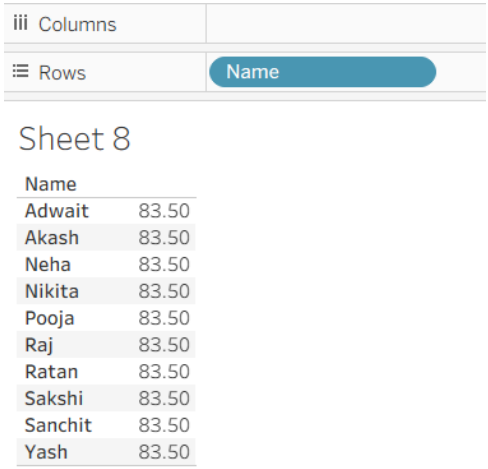
Now use EXCLUDE
{ EXCLUDE [Name] : AVG([Marks]) }
Here Tableau says:
"I know **Name** is present in the worksheet, but **ignore Name while calculating**."
So, Tableau temporarily changes this:
Department + Name
to
Department only

Now it calculates:
IT Department Average
(80+70+90)/3=80(80 + 70 + 90) / 3 = 80(80+70+90)/3=80
This value (80) is then shown for every student in the IT department.
Result:

Department	Name	EXCLUDE Result
IT	Amit	80
IT	Neha	80
IT	Ravi	80

Notice that the **student names are still visible** in the worksheet. **EXCLUDE does not remove the Name column from the view.** It only removes **Name from the calculation**.

{ EXCLUDE [Name] : AVG([Marks]) }



(F)Aim:- Create and Use Parameters for Dynamic Analysis.

A **Parameter** is **not connected to the data**. It is a user-controlled value. When the user changes the parameter, Tableau automatically updates any calculated fields that use it.
A **Parameter** is a dynamic value that allows users to interact with a Tableau visualization by changing values such as pass marks, discount percentage, year, or threshold.

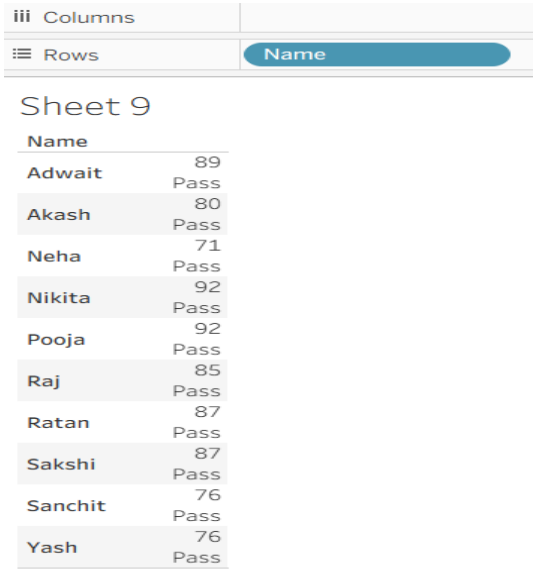
Step 1:
Right-click in the Data pane.
Choose: Create Parameter

Step 2:
Create:
Name: Pass Marks
Data Type: Integer
Current Value:40
Minimum:0
Maximum:100
Step Size:1
Click **OK**.

Step 3:
Create another Calculated Field.
Formula:
IF [Marks] >= [Pass Marks]
THEN "Pass"
ELSE "Fail"

END
Now changing the parameter automatically changes the pass/fail result.

Result :
Drag:
Name → Rows
Marks → Text
Result → Text.

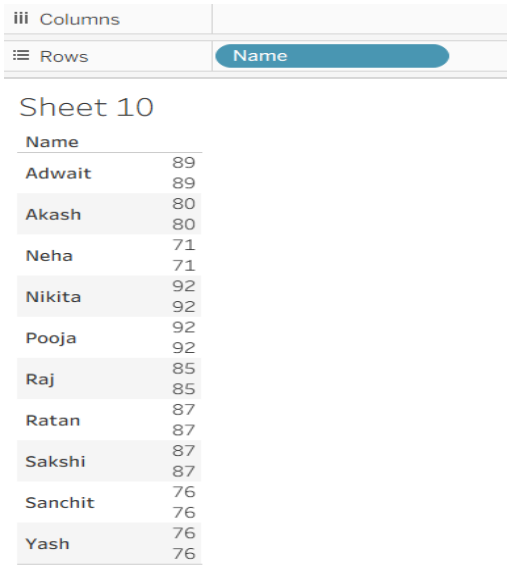


(G)Aim:- Apply Calculations to Fix Data Issues and Enhance Insights.

Example 1: Replace Null Values

IFNULL([Marks],0)
Drag:

- **Name** → Rows
- **Marks** → Text
- **Marks After IFNULL** → Text



Example 2: Attendance Category

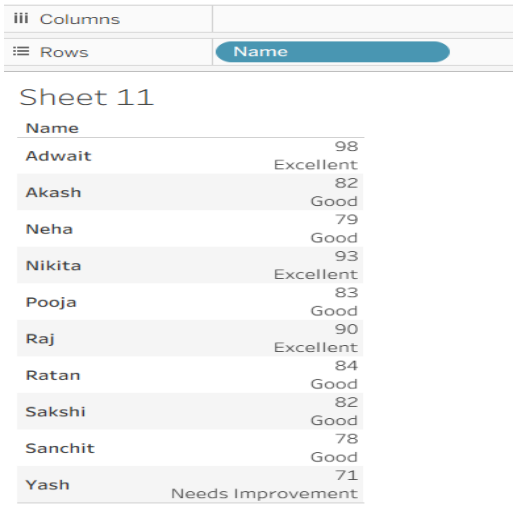
IF [Attendance] >= 90 THEN
"Excellent"

ELSEIF [Attendance] >= 75 THEN
"Good"

ELSE
"Needs Improvement"
END

Result :

Name → Rows
Attendance → Text
Attendance category → Text



Example 3: Grade

IF [Marks] >= 90 THEN "A"
ELSEIF [Marks] >= 75 THEN "B"
ELSEIF [Marks] >= 60 THEN "C"
ELSE
"Fail"
END

Result :
Name → Rows
Marks →Text
Grade → Text

